The Questions

1. For each part of this question, clearly state any assumptions you make.

a) Explain why small signal modelling and analysis is a useful tool in analogue electronics.

[3]

Small signal analysis simplifies the analysis of analogue circuits and presents the results in a simpler more useful way to the designer. It takes relationships that are exponential, or square law in equations that may be transcendental and impossible to solve without utilising numeric methods and linearizes the relationships. This means that over the range of operation of interest, a circuit can then often be described by a proportional relationship between input and output.

b) What is the operating mode of the MOSFET in Figure 1.1? What is the length of the inverted channel region compared to the distance between the drain and source diffusions? Explain why.

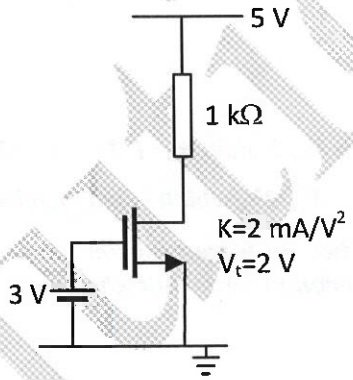


Figure 1.1

[6]

As the gate source voltage exceeds the threshold voltage, the MOSFET is on. We can first see if the MOSFET is in the saturation/active region by using the relationship:

$I_{Dsat}=k(V_{GS}-V_t)^2$.

This would give $I_{Dsat}=2.(3-2)^2=2mA$.

This puts the drain voltage at $(5-1k\Omega.2mA)=3$ *V.*

The requirement for saturation is that $V_{DS} > V_{GS}-V_t$, *so* $3>3-2$, *so we are indeed in saturation/active mode.*

In this mode the channel is pinched off and so the channel is shorted then the distance between the drain and source diffusions.

Solutions

c) Show that for the common emitter amplifier in Figure 1.2 the small gain is independent of the value of R_C assuming the quiescent output voltage is $V_{cc}/2$. You may neglect the transistor's output resistance. Suggest two ways of increasing the gain.

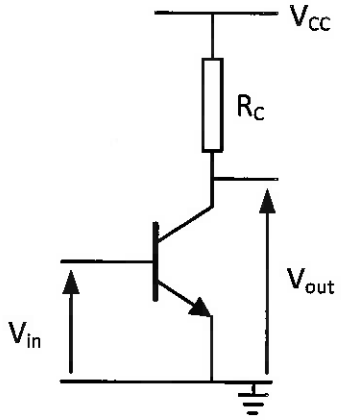


Figure 1.2

[6]

The small signal gain is:

$Av=-g_mR_c$

If the quiescent output voltage is $V_{cc}/2$, then $g_m=I_c/V_T= V_{cc}/(2R_cV_T)$

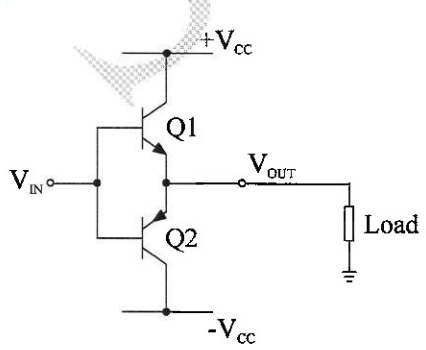
Making the gain $-V_{cc}R_c/(2R_cV_T)=-V_{cc}/2V_T$ which is independent of R_c.

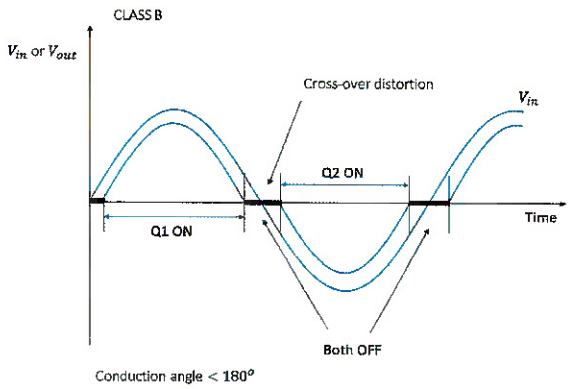
The gain can be increased either by increasing the power supply rail (which will increase quiescent power consumption) or by introducing an active load in place of R_c.

d) With the aid of diagrams, explain why the class B output stage suffers from cross-over distortion.

[4]

The class B stage is shown below and its output and input characteristics as a function of time are also shown.





The class B stage has a dead band on the input where neither the upper of lower transistor are on, giving rise to cross over distortion on the output as shown above.

Solutions

e) The circuits of Figures 1.3 and 1.4 are proposed for use as logic inverters. A logic low input is 0 V and a logic high input is 5 V. A logic high output is 5 V and a logic low output is a maximum of 0.2 V for both circuits.

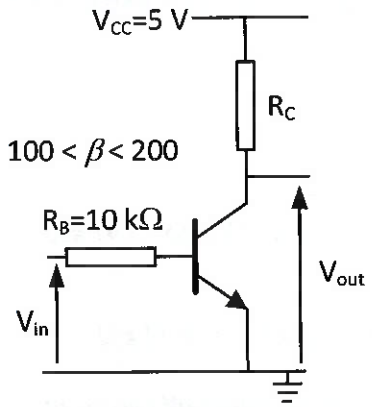


Figure 1.3

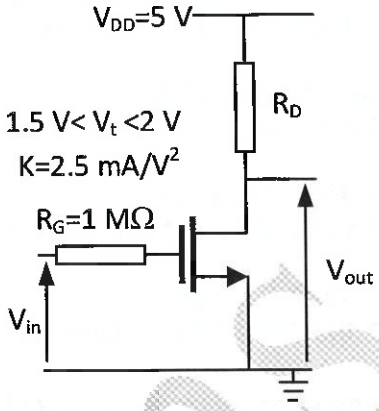


Figure 1.4

i) What is the value of R_c that causes the BJT to be overdriven by a factor of no more than 5 when the input voltage is 5 V?

[6]

*For an input of 5 V, I_B=(5-0.7)/10k=0.43mA. At an overdrive factor of 5, the forced beta varies between 100/5 and 200/5, i.e. 20 and 40 and hence collector currents vary between 0.43*20=8.6 mA and 0.43*40=17.2 mA.*

Assuming V_{CESAT}= 0.2V , this would set the respective values of R_c to: (5-0.2)/8.6m=558Ω for β=100 and (5-0.2)/17.2m=279Ω for β=200

If we choose R_c=558Ω when β=200, we end up with forced β of (8.6m/0.43m)=20 in the case that β is 200, corresponding to an overdrive factor of 10. If we choose R_c=279Ω for β=100, we have a forced β of (17.2m/0.43m)=40, which is an overdrive factor of 100/40=2.5.

Thus we choose Rc=279Ω

ii) Does the value of R_c calculated above allow the inverter to work to specification across the entire range of transistor tolerances?

[2]

Yes, the inverter will work over the entire range of tolerances of transistors if we choose R_c=279Ω. The value of β does not effect the performance when the BJT is off (i.e. a low input and a high output, which will always work assuming v_{in}<0.7V). For high inputs, when the BJT is on and the output is low, both values of β still overdrive the BJT by a factor of at least 2.5, causing the output to saturate at around 0.2 V, as per the specification.

iii) What is the minimum value of R_D that meets the specifications for the inverter for all possible MOSFETs? Ignore channel length modulation.

[4]

For a low input (guaranteed to be < 0.7 V), the inverter output voltage will always be high because we have no reached the threshold voltage of the MOSFET. In the case that the input is high, we must ensure that the output voltage of the inverter is <0.2 V.

In this case the MOSFET will be in the linear or triode region (V_{DS}=0.2 V which is less than (V_{GS}-V_t)=5-2V in the worst case).

We can therefore write:
$I_D=k(2(V_{GS}-V_t)V_{DS}-V_{DS}^2)$

When V_t=1.5 V, we get:
$I_D=2.5((5-1.5)0.2-0.2^2)$=1.65 mA, corresponding to a resistor of 4.8/1.65=2.91 kΩ

When Vt=2 V, we get
$I_D=2.5((5-2)0.2-0.2^2)$=1.4 mA, corresponding to a resistor of 4.8/1.4=3.43 kΩ.

Thus we choose a resistor of 3.43 kΩ. minimum that will work across all ranges of Vt.

f) Describe the advantages and disadvantages of each of the BJT amplifier configurations shown below in Figure 1.5(a-c). Include in your answer a comparison between their small signal gain, bias stability, and input resistance. There is no need to derive mathematical formulae.

[9]

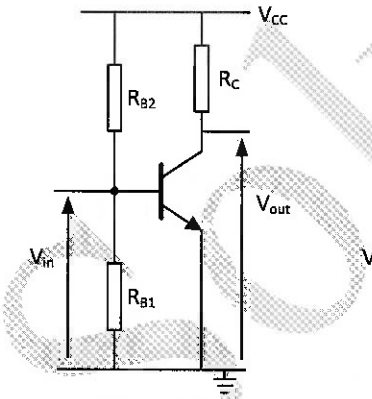


Figure 1.5a

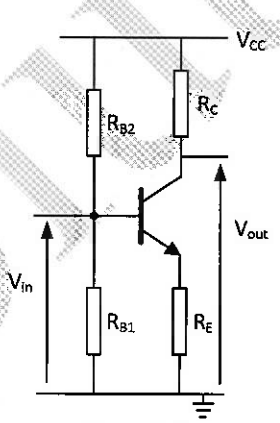


Figure 1.5b

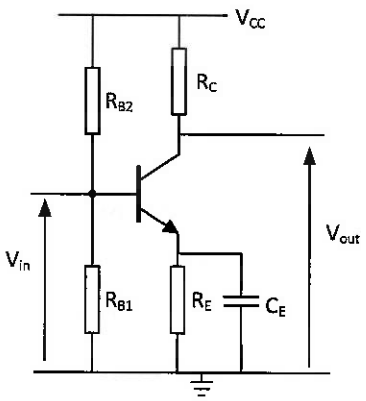


Figure 1.5c

The arrangement in 1.5a is the simplest scheme but suffers from bias stability issues, either through different beta values of the transistors (which is not a very well controlled value) or changes in gm/beta values due to temperature variation. The advantage of this circuit is that the gain is high, although the gain is temperatire dependent because it relies on g_m.

The arrangement in 1.5b is has significantly improved bias characteristics because the bias current is now almost independent of beta, as is the gain of the circuit. The problem with this circuit is the gain is significantly less than 1.5a because it reduces to $\alpha R_C/R_E$.

The final arrangement in 1.5c takes the best of both of these arrangements. At DC, for bias purposes, the circuit has the stability of 1.5b (i.e. doesn't rely on beta) and assuming that C_E is large enough, the snall signal gain is as high as 1.5a.

2. Figure 2.1 shows a single stage common emitter amplifier with an active load constructed from a simple current mirror. All transistors are matched and V_{CC} is 8 V.

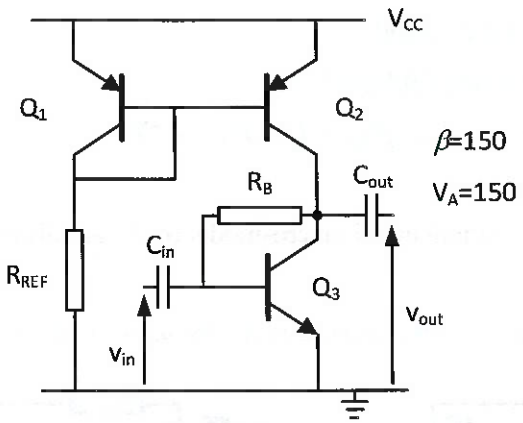


Figure 2.1

a) Neglecting finite beta error, choose a value of R_{REF} to bias the collector current of Q_1 at 1 mA and then choose a value of R_B that biases Q_3 to give a quiescent output voltage to give maximum head- and leg-room for the amplifier.

[6]

R_{REF} must be (8-0.7)/1m=7.3 kΩ to give a 1mA reference to the simple current mirror.

R_B is chosen as follows:

Maximum head- and leg-room requirements means to bias v_{out} at 4 V under quiescent conditions, thus:

$(\beta+1)(v_{out}-0.7)/R_B=I_{CQ2}=1mA$

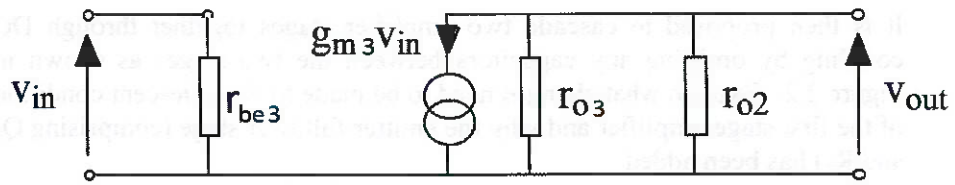
For v_{out}= 4V, we have:

R_B=498.3 kΩ

b) Draw a small signal model of the amplifier, and assuming that at the frequency of operation C_{in} and C_{out} are short circuits, calculate the input and output impedance of the amplifier, and its small signal gain. You may assume R_B is very large.

[8]

Neglecting R_B, The small signal model is as follows:



Where r_{o3} is the small signal output resistance of Q_3 and r_{o2} is the small signal output resistance of Q_2.

The small signal gain is therefore $-g_{m3}*(r_{o3}//r_{o2})$.

solutions

$r_{o3}=r_{o2}=V_A/I_C=150/1m=150k\Omega$

$g_{m3}=I_C/V_T=1m/25m=40$ mS

*Giving a gain of -40*150/2=-3000.*

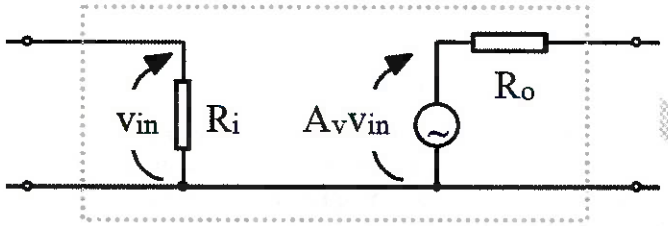
The output impedance is just 150k/2=75 kΩ

The input impedance is just $R_{BE}=\beta/g_{m3}=150/40m=3.75k\Omega$

c) Draw a labelled small-signal macro-model of the amplifier.

[4]

We have already calculated the parameters for the micromodel, and hence we have:



Where Ri=3.75kΩ, A_v=-3000 and R_O=75 kΩ.

d) It is proposed to link 2 of these amplifiers together with AC coupling to achieve a higher gain. Using the macro model from part c), what is the gain of the resulting circuit when driving a 10 kΩ load from a perfect voltage source? Assume all capacitors are short circuits at the operating frequency. Comment on your answer.

[6]

When the two amplifiers are connected, and assuming any coupling capacitance is a short circuit, the gain is significantly less than one might originally expect because the input of the second stage significantly loads the output of the previous stage (the input impedance of this amplifier is much less than its output impedance)

The circuit gain (not including effects from an input source impedance or a load impedance) is therefore:

*3000*3.75/(3.75+75)*3000*10/(10+75)=-50,420.*

This gain is very high, but significantly less than the product of the gain of the two stages because of the loading of one amplifier on another and due to the load.

e) It is then proposed to cascade two amplifier stages together through DC coupling by omitting any capacitors between the two stages as shown in Figure 2.2. Explain what changes need to be made to the quiescent condition of the first stage amplifier and why the emitter follower stage (comprising Q_7 and R_E) has been added.

[6]

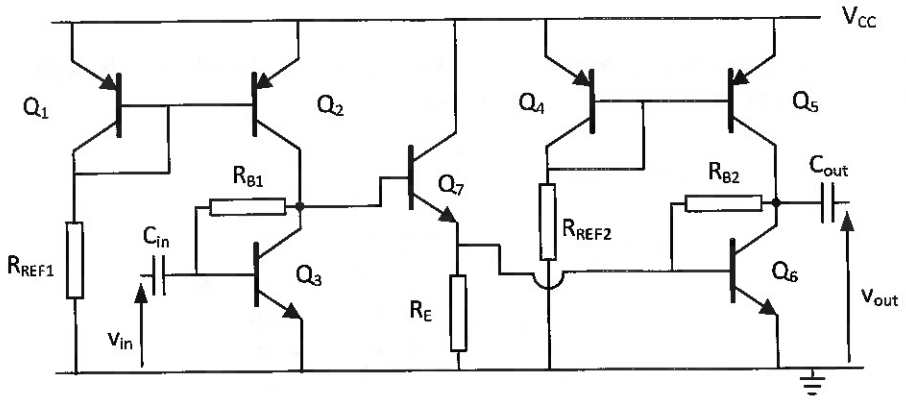


Figure 2.2

When cascading the two stages with DC coupling, we need to make sure the quiescent output of the first stage is appropriate to bias the second stage. If we bias the output of the first stage at Vcc/2, i.e. 4 V, then the output of the first stage is far too high to bias the second stage. Hence, we need to bias the first stage output to a much lower value, close to 0.7 V ready to drive the second stage.

However, if we bias the output of the first stage to 0.7 V to drive the second stage, there is little leg room for the output swing of stage 1. The output of the first stage should always stay above 0.7 V to keep the transistor in the forward active mode.

Hence, a possible solution is to add an emitter follower to allow the output of the first stage to sit at a quiescent value of 1.4 V, thus keeping the first stage in the forward active mode and allowing the second stage tobe properly biased at around 0.7 V on the base.

Solutions

3. Figure 3.1 shows a differential amplifier constructed using MOSFETs. Throughout this question assume all MOSFETs are matched.

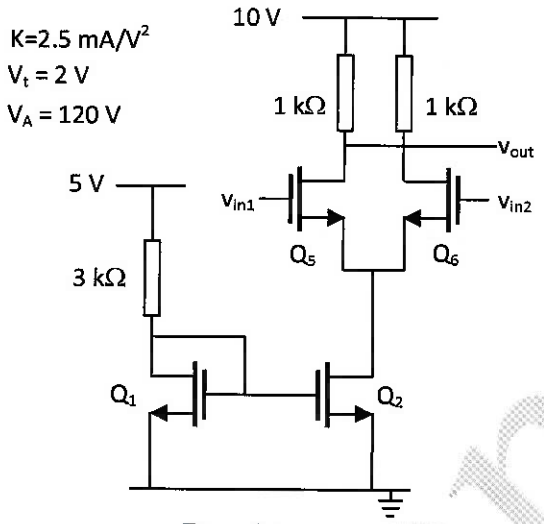


Figure 3.1

a)

i) Calculate the tail current set up by the current mirror comprising Q_1 and Q_2.

[4]

The tail current is produced by the simple current mirror, and the current in Q_1 is:

$I_D=K(V_{GS}-V_t)^2$, so:

(the device is in the saturation region)

$(5-V_{GS})/3k=2.5*(V_{GS}-2)^2$

$V_{GS}=2.57$ V (or 1.53 V as the other solution to the quadratic, which we must reject as it is below the threshold voltage)

And so the tail current $I_D=2.5*(0.57)^2=0.812$ mA

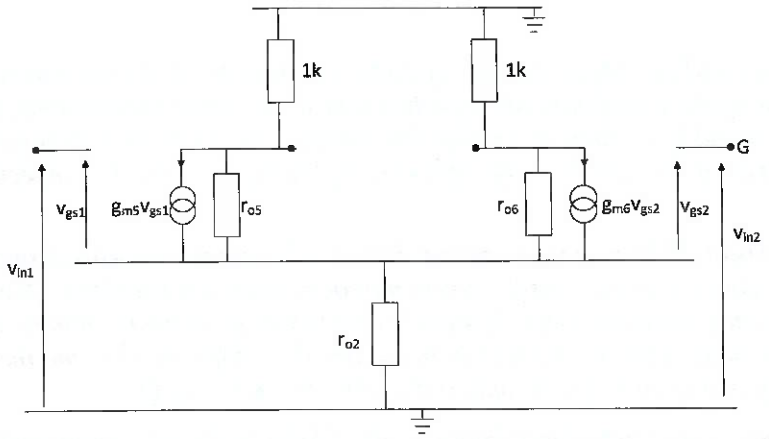
This means that half the current flows in each of Q_5 and Q_6, i.e. 0.406 mA

ii) Draw a small signal model of the circuit and calculate the small signal differential gain for a single-ended output (where $v_{in}=v_{in1}-v_{in2}$).

[10]

Small signal model is as follows:

Solutions



Due to symmetry, when a pure differential gain is applied, the source voltage of both devices is a virtual ground, and this then simplifies the analysis, because each side can be treated as a separate common drain amplifier. The single ended differential gain is therefore:

A_d=-gm(R_D//r_{o5})/2 (Note that the gain is negative because of where the output is taken)

*g_m for the MOSFET Q5 is 2*sqrt(k*I_D)= 2*sqrt(2.5*I_D)=2.01 mS.*

r_{o5}=V_A/I_D= 120/0.406=295.6 kΩ

Thus the differnetial gain is -2.01 m(1//295.6)/2=-2.01*(0.997)/2=-1*

b) By assuming changes in V_{GS} of Q_5 and Q_6 are negligible, approximate the common mode gain and calculate the common mode rejection ratio (CMRR).

[6]

If we assume that V_{GS} of Q_5 and Q_6 stay constant when we apply a common mode input, the common mode gain is simply -R_D/(2r_{o2})

r_{o2}=120/0.812m=147.8kΩ

*Thus the common mode gain is -1/(2*147.8)=-0.0034*

Thus the CMRR is 1/0.0034=294.1

c) It is proposed to improve the CMRR by replacing the simple MOSFET current mirror with a 3 transistor Wilson current mirror of Figure 3.2a. Explain why the output impedance of this current mirror is superior to that of the simple current mirror and why this improves the CMRR. There is no need to draw a small signal model or derive any equations.

[6]

SOLUTIONS

The reference and the output currents of even the simple MOSFET current mirror will be identical of the transistors are matched and at the same temperature, and have identical V_{GS} and V_{DS}. Their V_{GS} values are forced to be equal from their connection, but the values of V_{DS} can be very different, depending on what is connected to the output.

The 3 transistor Wilson current mirror has a substantially improved small signal output impedance over the simple current mirror because the transistor Q_3 effectively compresses the potentially large changes in output voltage to small changes in its V_{GS}. V_{GS} of Q_3 is small anyway and thus both the absolute difference in, and the changes in, V_{DS} of Q_1 and Q_2 are significantly reduced by the action of Q_3.

This improved output impedance improves the CMRR of the differential pair because it reduces the common mode gain. Common mode gain occurs because an increase in common mode input voltage tends to increase the voltage on the output side of the current mirror, increasing the tail current, causing a change in the output voltage.

d) The 4-transsitor version of the Wilson current mirror is shown in Figure 3.2b. From your knowledge of the characteristic curves of a MOSFET, explain why this current mirror achieves current gain closer to 1 than the 3 transistor version. Again, there is no need to draw an SSEM or derive any equations.

[4]

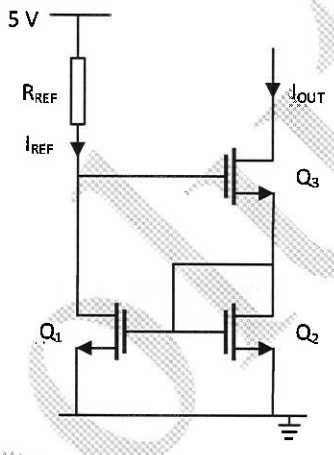


Figure 3.2a

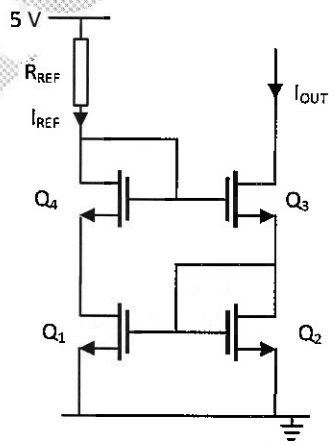


Figure 3.2b

The three transistor mirror substantially increases the output impedance of the current mirror because it substantially reduces the differences in V_{DS} between Q_1 and Q_2. However, the value of V_{DS} still differs by the value of V_{GS}. By adding the additional transistor, Q_4, the values of V_{DS} for Q_1 and Q_2 are now almost identical, meaning the mirror's current gain is now very close to 1. With reference to the characteristic curves of the MOSFET, the 3 transistor mirror operates Q_1 and Q_2 on the line of the same V_{GS} but with different values of V_{DS}. The current then differ because of channel length modulation/finite output impedance. For the 4 transistor version, Q_1 and Q_2 operate at exactly the same point on that line.

Solutions